AWESOME CYBER SECURITY HEAVEN — REALM DOSSIER · 2026 EDITION

The Proving Grounds

CTF Practice, Labs & Hacking Training

104	124,893	38%	18
FREE TOOLS	COMMUNITY STARS	ON GITHUB ACTIVE	ACTIVE IN 2026

104 tools · Part of Awesome Cyber Security Heaven — 1435 curated free tools across 14 realms · Data verified 2026-11-06 (re-verified monthly)

CURATOR'S NOTES

Legal targets to practice on — vulnerable-by-design apps, CTF platforms, lab ranges. Practice targets are verified live before listing (dead practice apps are worse than none — they teach tool errors, not vulns).

MAINTENANCE SNAPSHOT (2026-11-06)

- **Active in the last 30 days:** 17 of 47 tracked GitHub repos (36%)
- **Quiet classics (no push in 12+ months, still listed):** Metasploitable 3, HEVD, APTSimulator, OWASP NodeGoat, DIVA Android, DVIA-v2, AzureGoat, 0l4bs, WackoPicko, lfi-labs, DVHMA, CryptOMG, damnvulnerable.me, SentinelTestbed
- **Earliest entries:** WackoPicko (2010), CryptOMG (2012), pwntools (2013)

FLAGSHIP PICKS — THE HEAVYWEIGHTS

TOOL	★ STARS	CREATED	LAST PUSH	WHAT IT DOES
Vulhub	21.4k	2017	2026-10-17 ✅	Pre-Built Vulnerable Environments Based on Docker-Compose
OWASP Juice Shop	14.3k	2014	2026-10-30 ✅	OWASP Juice Shop: Probably the most modern and sophisticated insecure web application
DVWA	13.9k	2013	2026-10-18 ✅	Damn Vulnerable Web Application (DVWA)
pwntools	13.8k	2013	2026-10-30 ✅	CTF framework and exploit development library
WebGoat	9.4k	2015	2026-10-29 ✅	WebGoat is a deliberately insecure application
GOAD	8.6k	2021	2026-10-23 ✅	game of active directory

Full inventory — 104 curated tools · ranked by GitHub adoption · 2026-11-06 census · non-GitHub tools listed without stars

TOOL	★ STARS	CREATED	LAST PUSH	LICENSE / TIER
Vulhub	21.4k	2017	2026-10-17 ✅	— dockerized vulnerable environments: spin up a CVE, test, tear down. Used by learners spinning up vulnerable environments with Docker; the free compose-file library of real CVE setups. (MIT)
OWASP Juice Shop	14.3k	2014	2026-10-30 ✅	— the modern deliberately-vulnerable app with gamified challenges covering the whole OWASP spectrum. Used in corporate trainings and university courses; the de facto standard vulnerable app for web security practice.
DVWA	13.9k	2013	2026-10-18 ✅	— the classic Damn Vulnerable Web Application for safely learning injection basics. Used by instructors teaching web vulnerabilities for years; the classic PHP vulnerable app in every classroom lab. (GPL-3.0)
pwntools	13.8k	2013	2026-10-30 ✅	— the CTF exploit-development library: from sockets to ROP chains, pwn everything. Used by CTF players and exploit developers writing exploits fast; the standard Python library for pwn challenges.
WebGoat	9.4k	2015	2026-10-29 ✅	— OWASP's teaching app: guided lessons with a solution walkthrough for each vulnerability class. Used by Java teams and instructors learning web exploitation; the OWASP teaching app with lessons built in. (GPL-2.0)
GOAD	8.6k	2021	2026-10-23 ✅	— Game of Active Directory: a three-forest lab purpose-built for safely practicing AD attack techniques. Used by AD red-team trainees practicing real attack paths; the free Game of Active Directory lab commercial courses emulate. (GPL-3.0)

CTFd	6.9k	2015	2026-10-30 ✅	— the CTF platform that powers thousands of events; self-host your own competition. Used by CTF organizers self-hosting competitions; the open-source platform behind thousands of events.
Metasploitable 3	5.8k	2016	2025-02-13 ⏳	— Rapid7's purpose-built vulnerable target VM; the canonical Metasploit practice range. Used by pentest trainees practicing Metasploit workflows; the Rapid7-blessed free practice target.
CloudGoat	3.8k	2018	2026-10-14 ✅	— Rhino Security's "vulnerable by design" AWS scenarios — deploy, attack, tear down. Used by cloud security learners practicing AWS attacks safely; Rhino Security's free vulnerable-by-design deployment.
HEVD	3.1k	2015	2025-02-24 ⏳	— the Windows kernel-driver training target for exploit development. Used by kernel exploit developers learning Windows driver exploitation; the free HackSys training driver with documented vulns. (GPL-3.0)
H5SC	2.9k	2014	2026-09-09 ⏳	— A collection of HTML5-based XSS test vectors and cheat sheets. Used by security testers; a practical payload reference for XSS testing. (MPL-2.0)
APTSimulator	2.8k	2018	2025-09-23 ⏳	— makes a Windows host look compromised so you can test detections safely. Used by blue teams generating APT-style artifacts for detection testing; the free Windows simulator for purple exercises. (MIT)
Splunk Attack Range	2.6k	2019	2026-10-26 ✅	— builds instrumented detection labs: attack, collect and test in one workflow. Used by detection engineers building full attack/defend labs in the cloud; the free Splunk range that emulates commercial cyber ranges. (Apache-2.0)
OWASP NodeGoat	2.1k	2013	2024-06-15 ⏳	— the Node.js vulnerable-by-design app: the Top 10 in a controllable lab. Used by Node.js teams learning the OWASP Top 10 in their stack; the OWASP training app for JavaScript. (Apache-2.0)
OWASP Mutillidae II	1.5k	2018	2026-09-24	— deliberately vulnerable app that re-creates every OWASP Top 10 issue, resettable per install. Used by teachers demonstrating nearly every OWASP vulnerability; the deliberately vulnerable training app. (GPL-3.0)
DIVA Android	1.2k	2016	2023-05-19 ⏳	— Damn Insecure and Vulnerable App: the Android training classic. Used by mobile security students learning Android flaw classes; the classic deliberately vulnerable Android app. (GPL-3.0)
DVIA-v2	1.1k	2018	2024-03-29 ⏳	— Damn Vulnerable iOS App v2: the mobile testing curriculum as an app. Used by iOS security learners practicing on Swift; the free deliberately vulnerable iOS app. (MIT)
Invoke-AtomicRedTeam	1.1k	2020	2025-09-08 ⏳	— PowerShell module for executing Atomic Red Team tests, automating reproducible execution of 1,000+ ATT&CK techniques. Used by purple teams and detection engineers; the standard ATT&CK test runner. (MIT)
UERANSIM	1.1k	2019	2026-10-14 ✅	— Open-source 5G UE and RAN simulator commonly used for fuzzing NAS and RRC protocols in telecom security research. Used by telecom researchers; a standard testbed component. (AGPL-3.0)
ICSim (Instrument Cluster Simulat…	1.0k	2014	2025-09-18 ⏳	— CAN bus simulator emulating a vehicle instrument cluster for safe, legal testing of CAN injection, replay, and reverse engineering techniques. Used by automotive security researchers; a practical lab for learning vehicle bus attacks. (GPL-3.0)
AzureGoat	969	2022	2024-10-30 ⏳	— the vulnerable-by-design Azure cloud lab. Used by cloud security learners practicing Azure attacks; the free vulnerable-by-design Azure lab. (MIT)
DSVW	884	2015	2026-08-19	— Damn Small Vulnerable Web: the Top 10 in ~100 lines of Python. Used by trainers demonstrating web flaws with a single Python file; the deliberately small vulnerable app. (public domain)
SamuraiWTF	576	2016	2026-07-27 ⏳	— Web application penetration testing environment for practice. Used by pentesters and learners; a hands-on lab for web app testing. (LGPL-3.0)
Vulnerable Node	493	2016	2026-10-06 ✅	— the Node.js vulnerable application for testing analysis tools. Used by Node.js teams auditing a realistic vulnerable app; the free Express-based training target. (MIT-style)
TTPForge	447	2022	2026-09-30 ⏳	— Meta's cyber attack simulation platform built by red, blue, and purple teams to automate attacker TTPs. Used by purple teams and detection engineers; an open-source adversary emulation framework. (MIT)
DVWS (Web Sockets)	364	2017	2026-07-24	— the vulnerable WebSocket app from Interference Security. Used by students practicing WebSocket-specific attacks; the vulnerable websockets lab. (MIT)

0l4bs	355	2020	2021-06-02 ⏳	— XSS practice labs with progressive difficulty. Used by students practicing SQLi, XSS and command injection individually; the free isolated labs per bug class. (GPL-3.0)
WackoPicko	354	2010	2024-05-25 ⏳	— the vulnerable app built for testing vulnerability scanners — a frozen-by-design calibration target for your tooling. Used by researchers and instructors demonstrating classic web flaws; the free academic vulnerable social network.
lfi-labs	335	2015	2024-04-08 ⏳	— LFI/RFI/commix practice pages: one flaw family, many variants. Used by learners practicing local file inclusion safely; the free PHP lab for LFI classes. (MIT)
GRFICSv3	284	2025	2026-10-20 ✅	— Free and open-source OT security lab with realistic networking and 3D process simulation, featuring PLCs, HMIs, engineering workstations, and Kali. Used by ICS defenders and students; a safe, realistic practice range for industrial attacks. (GPL-3.0)
Java Vulnerable Lab	278	2015	2026-10-09 ✅	— the Java vulnerable web application from the CSPF crew. Used by Java developers learning exploitable patterns; the free vulnerable Java application. (GPL-2.0)
DVHMA	273	2015	2018-08-22 ⏳	— the Damn Vulnerable Hybrid Mobile App: Android+iOS appsec in one codebase. Used by mobile security students practicing hybrid-app flaws; the vulnerable hybrid mobile app lab. (Apache-2.0)
PureEdgeSim	220	2018	2026-11-01 ✅	— Simulation tool for edge and IoT computing environments used to evaluate security models and attack scenarios. Used by academic researchers; a flexible modeling framework. (GPL-3.0)
CryptOMG	195	2012	2015-06-25 ⏳	— cryptographic flaw testbed from SpiderLabs. Used by crypto learners attacking flawed implementations; the free crypto-focused challenge set. (GPL-3.0)
LearnPwn	191	2021	2026-09-04 ⏳	— Sample problems and solutions for learning binary exploitation with progressive difficulty. Used by security students and CTF players; a hands-on practice resource.
VWGen	85	2016	2017-12-10 ⏳	— Generator of vulnerable web applications for practice. Used by security learners and trainers; a handy lab-building utility.
OopsSec Store	55	2025	2026-11-02 ✅	— Intentionally vulnerable e-commerce application built with Next.js for practicing web security. Used by security learners; a deliberately insecure training target. (MIT)
Cooja Simulator	38	2017	2026-11-05 ✅	— Contiki OS network simulator for emulating resource-constrained IoT devices and implementing attack scenarios. Used by IoT researchers; a standard simulation testbed. (BSD-3-Clause)
damnvulnerable.me	34	2015	2019-05-19 ⏳	— a modern vulnerable webapp for the OWASP Top 10 era. Used by mobile security students learning both Android and iOS flaws; the free vulnerable mobile apps. (MIT)
XSS Labs	22	2026	2026-06-23 ⏳	— XSS lab exercises for learning web application security. Used by learners; a hands-on environment for practicing cross-site scripting. (NOASSERTION)
SimRun	13	2026	2026-11-05 ✅	— IBM's attack simulation platform for detection engineering that detonates attacks and verifies expected SIEM alerts fire. Used by detection engineers and purple teams; a practical detection-validation framework. (Apache-2.0)
SentinelTestbed	11	2013	2016-11-18 ⏳	— a testbed app for evaluating security instrumentation. Used by engineers testing detection content against a deliberately vulnerable host; the free detection-lab target. (GPL-3.0)
Vulnerable Login	6	2026	2026-07-28 ⏳	— A deliberately vulnerable login form built for security education and hands-on training. Used by learners and instructors; a simple practice target for authentication flaws. (MIT)
Open Redirect	4	2026	2026-07-28 ⏳	— An educational web application that demonstrates open redirect vulnerabilities in a safe lab setting. Used by students and trainers; a focused practice target for redirect flaws. (MIT)
Vulnerable Shopping Cart	3	2026	2026-07-28 ⏳	— Deliberately insecure e-commerce application for security practice. Used by learners and testers; a safe target for web exploitation exercises. (MIT)
AEVisionLab	0	2024	2026-04-03 ⏳	— Automotive Ethernet test platform using off-the-shelf ECUs for sniffing, interception, injection, and fuzzing. Used by vehicle security researchers; an open testbed.

IDS-Defender	0	2026	2026-04-18 ⏳	— Educational IDS/IPS deployment project using Snort 3 and Suricata on Kali Linux with Hydra attack simulation. Used by learners; a hands-on lab for practicing detection deployment. (MIT)
Altoro Mutual	—	—	—	— the classic demo bank: legal target for appsec training and tool demos. Used by security teams demoing scanners on a safe target; the classic IBM testfire-style demo banking site. (free service)
Arizona Cyber Warfare Range	—	—	—	— free live-fire ranges for defenders and researchers. Used by learners attacking live-fire lab environments; the free volunteer-run range. (free service)
bWAPP	—	—	—	— the "buggy web application" with 100+ vulnerability flavors — an old-school playground that still teaches. Used by instructors demonstrating 100+ web bug classes; the free buggy app for classroom labs.
Crackmes.one	—	—	—	— the reverse-engineering challenge community: thousands of crackmes, all legal. Used by reverse-engineering learners practicing on thousands of binaries; the free community crackme archive. (free service)
CryptoHack	—	—	—	— learn modern cryptography by breaking it: challenges from classical ciphers to ECC. Used by learners leveling up applied cryptography; the free interactive challenge platform.
Cryptopals	—	—	—	— the legendary crypto challenges; finish them and AES is no longer magic. Used by engineers learning real-world cryptography; the free challenge set that trained a generation of crypto implementers.
CTFlearn	—	—	—	— the beginner-friendly CTF platform with a permanent challenge set. Used by absolute beginners solving first flags; the free beginner-friendly CTF platform. (free service)
CTFtime	—	—	—	— the global CTF calendar, team rankings and writeup index — the hub of the sport. Used by CTF teams tracking events and rankings worldwide; the free calendar of every serious competition.
CyberDefenders	—	—	—	— blue-team CTF platform: investigate realistic incidents with the tools a SOC actually uses. Used by aspiring DFIR analysts practicing blue-team cases; the free DFIR CTF platform. (free tier)
ExploitMe Mobile	—	—	—	— the SecurityCompass mobile labs. Used by mobile security learners practicing iPhone and Android labs; the free mobile range. (free service)
Firing Range	—	—	—	— Google's XSS testbed: the original scanner benchmarking target. Used by XSS researchers testing payloads systematically; Google's free testbed for cross-site scripting classes. (Apache-2.0)
Gandalf	—	—	—	— Lakera's free prompt-injection game; learn AI-security attacks hands-on, level by level. Used by teams learning prompt injection hands-on; Lakera's free AI-security game.
Google Gruyere	—	—	—	— Google's cheese-themed vulnerable web app: exploit XSS, XSRF and information disclosure in the cloud. Used by beginners hunting their first web bugs; Google's free vulnerable-app tutorial.
Hack The Box	—	—	—	— the platform for ranked machines, pro labs and active-dir ranges; the industry's proving ground. Used by aspiring OSCP candidates and professional pentesters keeping sharp; the leading freemium training range. (free tier)
Hack This Site	—	—	—	— the community wargames veteran: missions, articles and forums since the early 2000s. Used by a generation of learners starting out; the veteran free training community.
Hackertest.net	—	—	—	— the 20-level hacking simulation for web basics. Used by beginners playing a level-by-level web puzzle; the veteran free challenge series. (free service)
HackThis	—	—	—	— 50+ levels of web security challenges, from basics to advanced. Used by beginners practicing across 50+ mission levels; the free UK challenge community. (free service)
Hackxor	—	—	—	— a realistic web-app hacking game with a story arc — WebGoat with sharper teeth. Used by aspiring web hackers playing a realistic first-person hacking game; the free browser-based range.
LetsDefend	—	—	—	— browser-based SOC simulator: alert queues, playbooks and IR challenges mapped to MITRE ATT&CK. Used by aspiring SOC analysts simulating alert triage; the freemium blue-team range with realistic cases. (free tier)

MASTG Crackmes	—	—	—	— the mobile RE challenges companion to the OWASP MASTG. Used by mobile security learners practicing the OWASP MASTG on real targets; the free crackme suite. (CC-BY-SA-4.0)
Microcorruption	—	—	—	— embedded-security CTF: reverse the lock firmware, level by level. Used by embedded security learners solving lock-firmware puzzles; the free embedded CTF from Matasano lineage. (free service)
MysteryTwister C3	—	—	—	— the international cryptography challenge: from Caesar to quantum. Used by crypto enthusiasts solving graded cipher challenges; the free crypto CTF with legendary pedigree. (free service)
OverTheWire	—	—	—	— Bandit and friends: learn Linux, networking and protocols through SSH wargames. Used by learners mastering Linux and security fundamentals; the classic free wargames every starter list recommends.
OWASP iGoat	—	—	—	— the iOS learning app from the OWASP mobile family. Used by iOS security learners practicing on Objective-C; the OWASP deliberately vulnerable iOS app. (wiki project)
Pentest-Ground	—	—	—	— hosted practice playground of intentionally vulnerable apps for security testing. Used by testers practicing against free lab environments; the training range from the Netsparker lineage. (free service)
picoCTF	—	—	—	— CMU's beginner-friendly CTF — the best first flag anyone can capture. Used by high-school and university students in CMU's free beginner CTF; the on-ramp for countless security careers.
pwnable.kr	—	—	—	— the classic system-hacking wargame: exploit your way from userland to kernel. Used by exploit developers practicing binary pwn; the classic free challenge server.
Reversing.kr	—	—	—	— the classic Korean reverse-engineering challenge site. Used by RE learners practicing Korean-style binary challenges; the classic free challenge site. (free service)
RingZer0 CTF	—	—	—	— hundreds of challenges across web, crypto, forensics and reversing. Used by learners solving web, crypto and coding challenges; the free online CTF platform. (free service)
Root Me	—	—	—	— hundreds of real-world hacking challenges across dozens of categories, from web to stego. Used by French-speaking and international learners; the free platform with hundreds of hands-on challenges. (free tier)
SmashTheStack	—	—	—	— the wargames network: shellcoding, exploitation and logic puzzles. Used by learners playing classic wargames like Amateria and Blowfish; the veteran free wargame community. (free service)
SocketToMe	—	—	—	— Robin Wood's WebSocket testing challenge app. Used by learners practicing network socket exploitation; the free deliberately vulnerable service. (free service)
techlogia Lab	—	—	—	— free browser-based hacking lab for security training. Used by students practicing end-to-end web exploitation; the free vulnerable lab environment. (CC-BY-4.0)
testasp.vulnweb.com	—	—	—	— the ASP demo target for scanner testing. Used by scanner developers and QA validating detection; Acunetix's free intentionally-vulnerable classic ASP test site. (free service)
testaspnet.vulnweb.com	—	—	—	— the ASP.NET demo target for scanner testing. Used by scanner developers and QA validating detection; Acunetix's free intentionally-vulnerable ASP.NET test site. (free service)
TryHackMe	—	—	—	— guided rooms from your first command to advanced red/blue paths — the friendliest on-ramp. Used by newcomers learning security step by step; the guided-labs platform that made entry-level training free. (free tier)
VulnHub	—	—	—	— downloadable vulnerable VMs; build and break your own home lab offline. Used by students building offline home labs; the free vulnerable-VM library that replaced expensive lab hardware.
Vulnserver	—	—	—	— the TCP vulnerable server for exploit-dev practice. Used by exploit developers practicing buffer overflows and ROP; the standard free TCP exploitation trainer. (free service)
W3Challs	—	—	—	— pentest training challenges with a ranking system. Used by learners practicing web, crypto and stego; the free challenge platform. (free service)

WeChall	—	—	—	— the challenge-site aggregator: one profile across all major wargames. Used by CTF players tracking ranks across challenge sites; the free aggregator for wargame rankings. (free service)
XSS Game	—	—	—	— Google's XSS tutorial game: six levels from context basics to DOM tricks. Used by beginners learning XSS exploitation; Google's free reverse-engineering-free XSS CTF. (free service)
Zero Bank	—	—	—	— demo banking site for web security exercises. Used by learners practicing web vulns on a simulated banking app; the free demo target behind Invicti training material. (free service)
pwn-scenarios	4	2026	2026-11-01 ✅	— 36k+ structured pentest/bug-bounty playbooks with expected observations per step, grounded in disclosed reports, auto-collected daily. (MIT + CC BY 4.0)
Open5GS	—	—	—	— Open-source project for emulating 4G and 5G core network functions, widely used as a test environment for security research and fuzzing. Used by telecom researchers; a standard lab core.
HTB SATCOM Security Essentials	—	—	—	— Hands-on CTF pack covering orbital mechanics, RF signal analysis, CCSDS protocols, DVB-S2 demodulation, and satellite firmware reverse engineering. Used by space security learners; a themed challenge set for SATCOM skills. (free tier)
ModbusPal	—	—	—	— Modbus slave simulator for creating virtual Modbus devices for security testing, protocol learning, and attack validation. Used by ICS testers and students; a convenient way to stand up realistic Modbus targets.
Stratus Red Team	—	—	—	— Datadog's cloud-focused attack simulation tool enabling granular detonation of offensive techniques in AWS and other clouds. Used by cloud security and purple teams; an open cloud attack emulation tool. (free tier)
SentinelX	—	—	—	— Modular CLI framework for authorized security testing, defensive validation, and purple team simulations on Kali Linux. Used by purple teamers and defenders; a scriptable simulation and validation toolkit.
Kali Autopilot	—	—	—	— Kali Linux tool for developing automatic attack scripts for red and purple teaming. Used by red and purple teamers; a scripted attack-automation utility in the Kali toolkit.
RangeForce Community	—	—	—	— Free cloud-based community SOC simulation labs using Splunk, Wireshark, and Suricata; used by defenders to practice hands-on skills. (free tier)
Hacker101 CTF	—	—	—	— HackerOne's free CTF platform with real-world web security challenges; used by aspiring bug hunters to practice vulnerability hunting. (free tier)
WebVerse Labs	—	—	—	— Hands-on hacking labs, challenges, learning paths, and web ranges. Used by learners and practitioners; a practice platform for offensive skills.
SELinux Game	—	—	—	— Interactive game for learning SELinux by solving puzzles and demonstrating skill. Used by Linux administrators; a hands-on SELinux practice platform.
FLAWS	—	—	—	— Amazon AWS CTF challenge covering cloud misconfigurations and privilege escalation. Used by cloud security learners; a well-known hands-on AWS practice range. (free, proprietary)
alert(1) to win	—	—	—	— Series of browser-based XSS challenges with escalating difficulty. Used by web security learners; a classic hands-on XSS practice site. (free, proprietary)
XSS Challenges	—	—	—	— Series of XSS challenges hosted by yamagata21. Used by web security learners; a long-standing hands-on XSS practice site. (free, proprietary)
HackSimulator	—	—	—	— A ChatGPT-based GPT acting as a hacking CTF that quizzes users on their experience. Used by learners and hobbyists; an interactive capture-the-flag style challenge. (free tier)